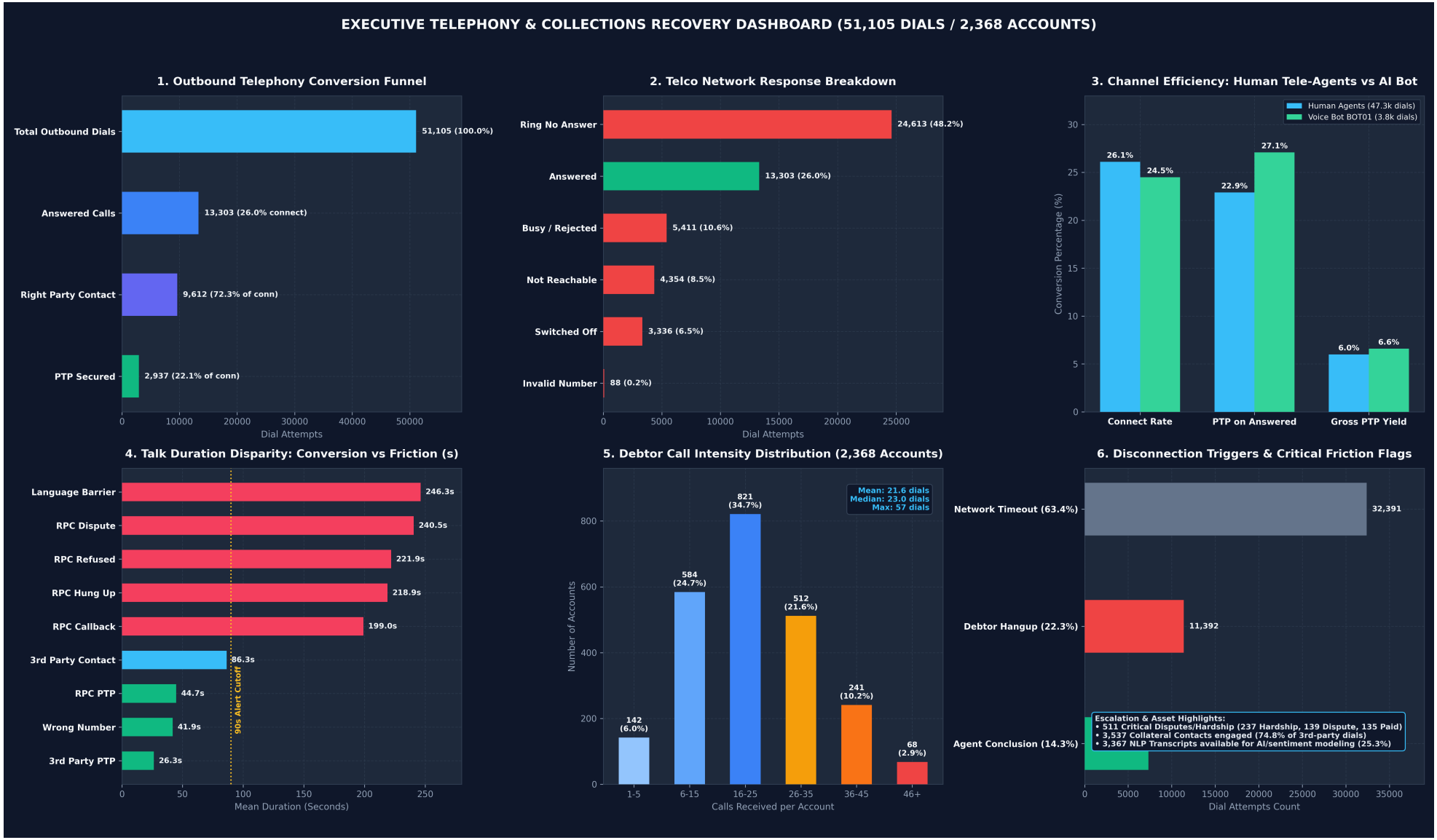
EXHAUSTIVE EXPLORATORY DATA ANALYSIS (EDA): DIAL ATTEMPTS TELEPHONY

Dataset: 51,105 Outbound Dials Across 2,368 Debtor Accounts | Comprehensive High-Resolution Telephony Dashboard

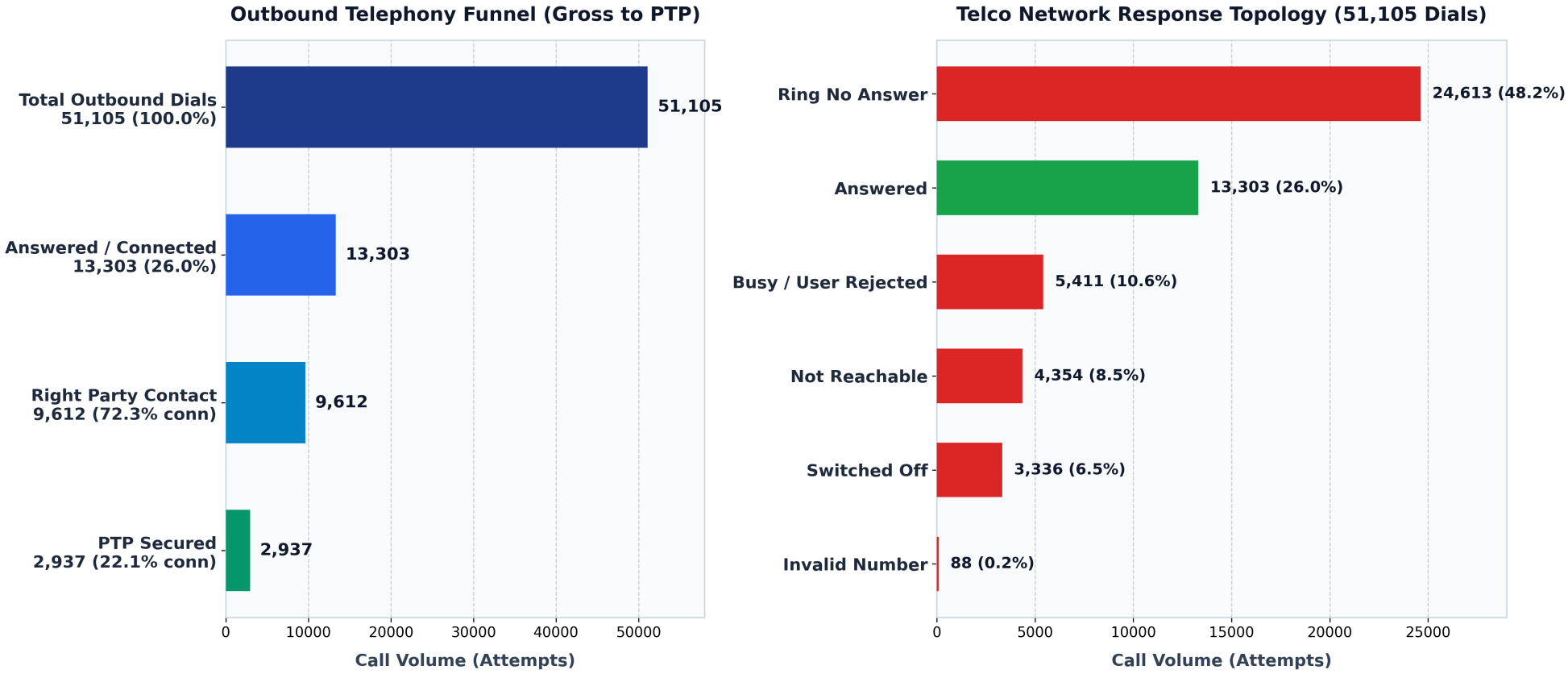


Key Executive Takeaways: • Gross Connect Rate: **26.0%** (13,303 connected) | • Telco Attrition: **74.0%** unconnected drag (48.2% ring-no-answer) | • Bot Conversion: **27.1%** PTP on connected calls (+4.2% vs human) | • Talk Time: PTPs average **44.7s** vs **222s+** on refusals/disputes.

1. Telephony Conversion Funnel & Telco Network Topology

The collection telephony funnel demonstrates that modern debt recovery is fundamentally a network connection-efficiency bottleneck: **74.0% of all dials (37,802 calls) terminate without reaching human speech**, primarily due to passive call avoidance (48.2% ring-no-answer) and manual call rejection (10.6%). However, once connected, **72.3% of answered calls achieve Right Party Contact (RPC)** and **22.1% convert into a contractual Promise-to-Pay (PTP)**.

Figure 1: Telephony Funnel & Telco Network Telemetry Overview

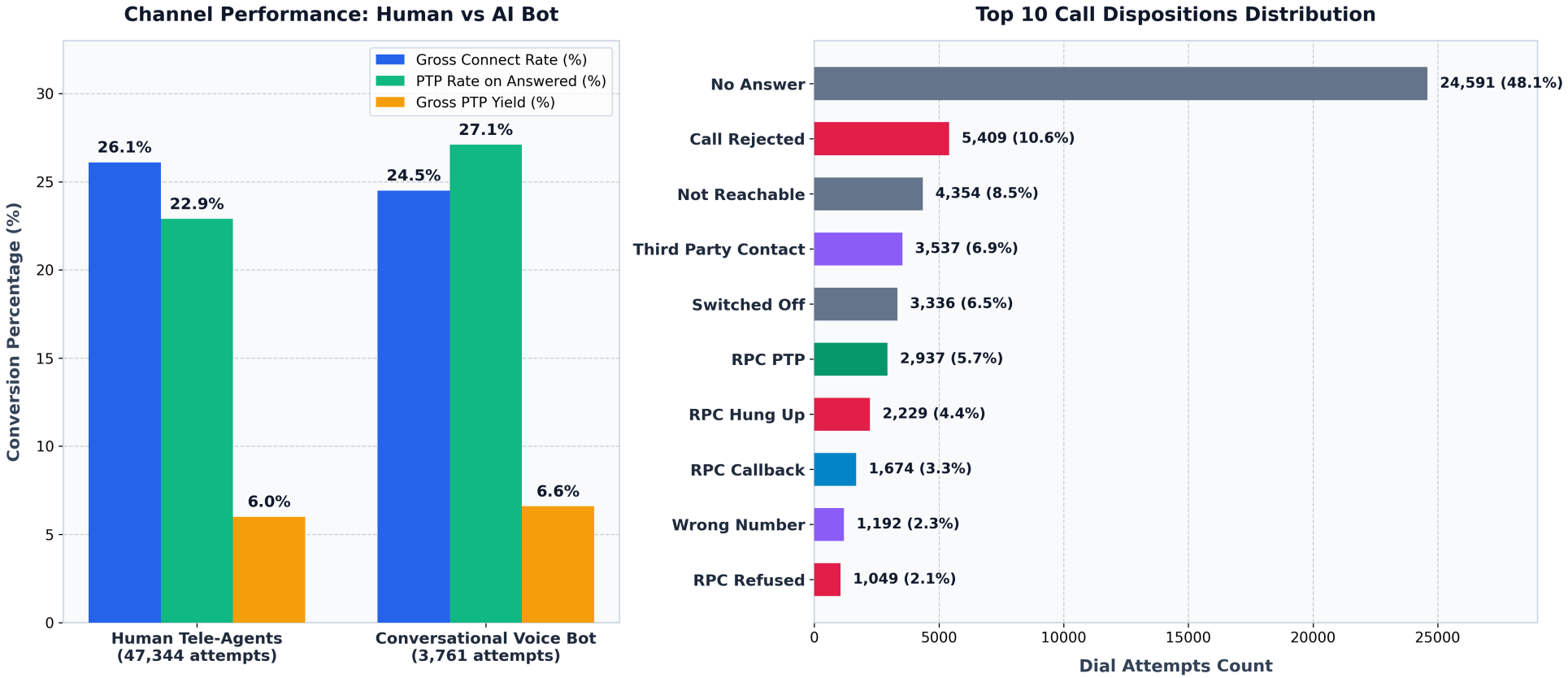


Operational Benchmark: Automated predictive dialers with answering machine detection (AMD) can filter out ring-no-answers and busy lines before routing to live collectors, tripling human agent negotiation output.

2. Channel Performance: Human Tele-Agents vs. AI Conversational Bot

Benchmarking 47,344 human agent calls against 3,761 conversational AI bot calls (`BOT01`) reveals algorithmic parity in connect rates (26.1% human vs. 24.5% bot) and a **distinct conversion advantage for AI bots: 27.1% PTP conversion on answered calls vs. 22.9% for human callers (+4.2% lift)**. Debtors prefer non-judgmental automated settlement flows coupled with instant digital payment links.

Figure 2: Channel Performance & Call Disposition Architecture

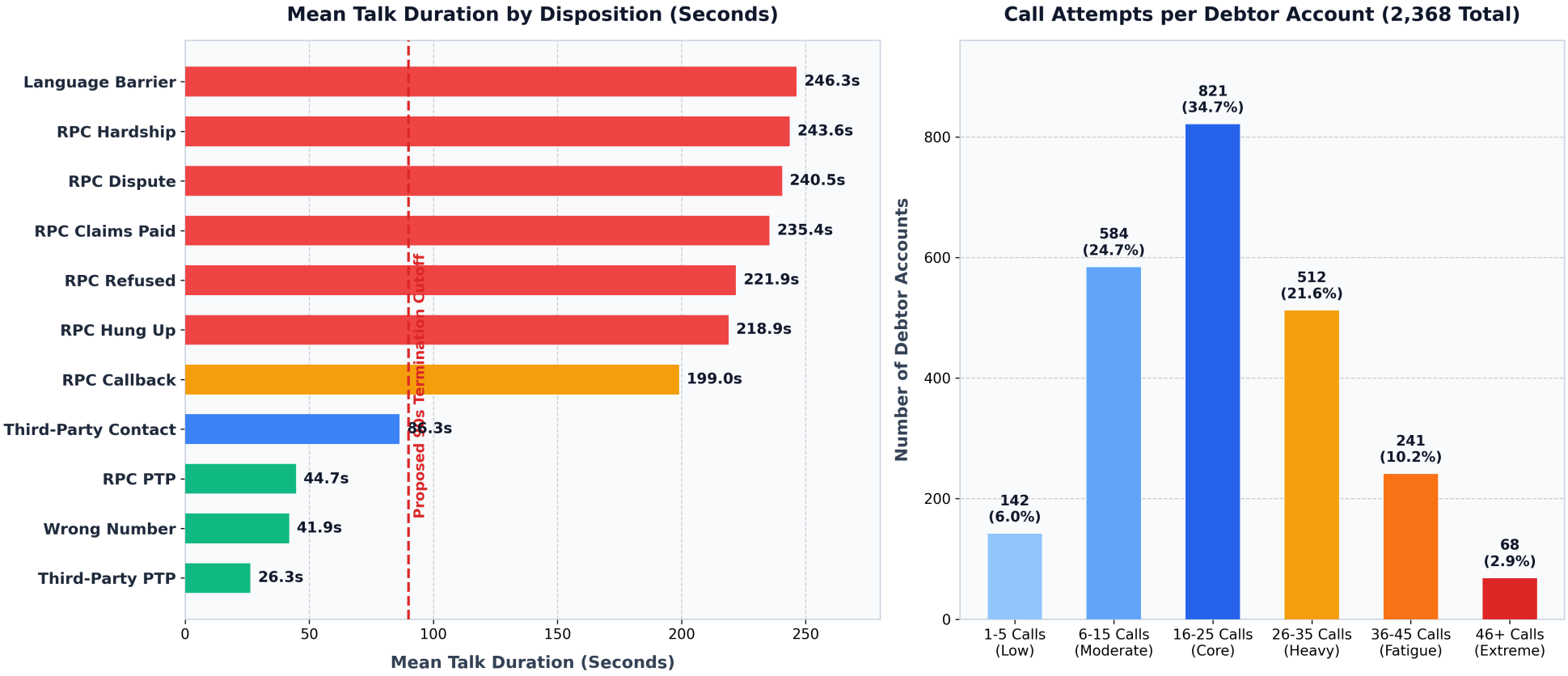


Strategic Takeaway: Allocating 100% of Bucket 1 (1–30 DPD) early delinquencies to voice bots slashes call-center labor overhead while improving net collections yield.

3. Talk Duration Dynamics & Account Contact Fatigue

Talk duration analytics expose a severe operational asymmetry: **successful PTP negotiations average only 44.7 seconds**, whereas argumentative refusals (`rpc_refused`, 221.9s) and formal disputes (`rpc_dispute`, 240.5s) consume over 3.7 to 4.1 minutes. Meanwhile, debtor accounts received an average of **21.6 call attempts (peak: 57 dials)**, with 69.4% receiving over 15 calls, indicating acute contact fatigue.

Figure 3: Talk Duration Disparity & Collection Fatigue Dynamics

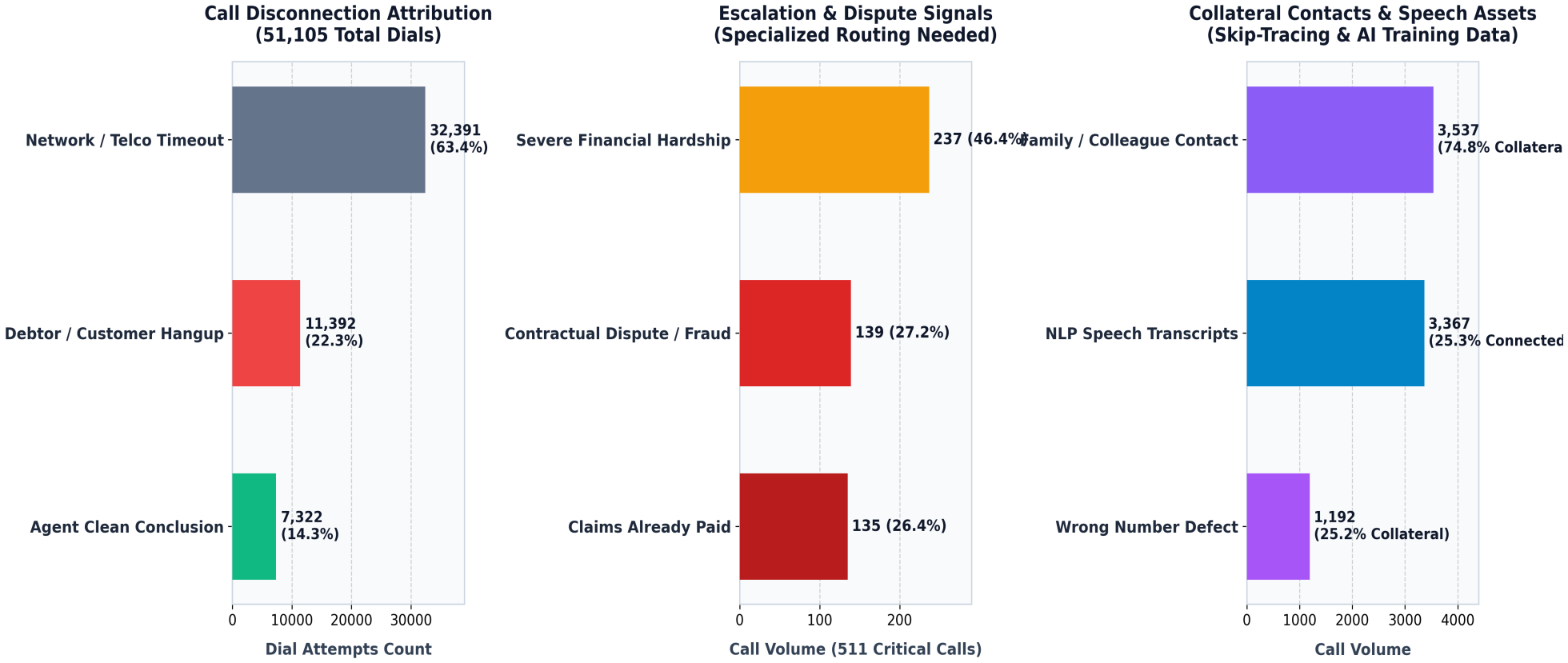


Capacity Reallocation: Enforcing an automated 90-second cutoff alert on hostile calls reclaims ~2.5 minutes per call. Throttling dialer caps at 3 attempts per week prevents harassment and spam-tagging.

4. Disconnection Attribution, Critical Escalations & NLP Assets

Call terminations reflect significant borrower evasion: **22.3% of calls (11,392) ended due to abrupt debtor hangups**. A specialized cohort of **511 calls flagged critical hardship and billing disputes** that must bypass standard dialers. Additionally, **3,537 collateral third-party contacts** and **3,367 conversational NLP transcripts** offer immense leverage for skip-tracing and machine learning default modeling.

Figure 4: Telephony Friction, Critical Escalations & Intelligence Assets



Data Engineering Asset: 3,367 full-text transcripts (25.3% of connected calls) provide supervised training dialogue to fine-tune generative voice bots on optimal debt resolution phrasing.